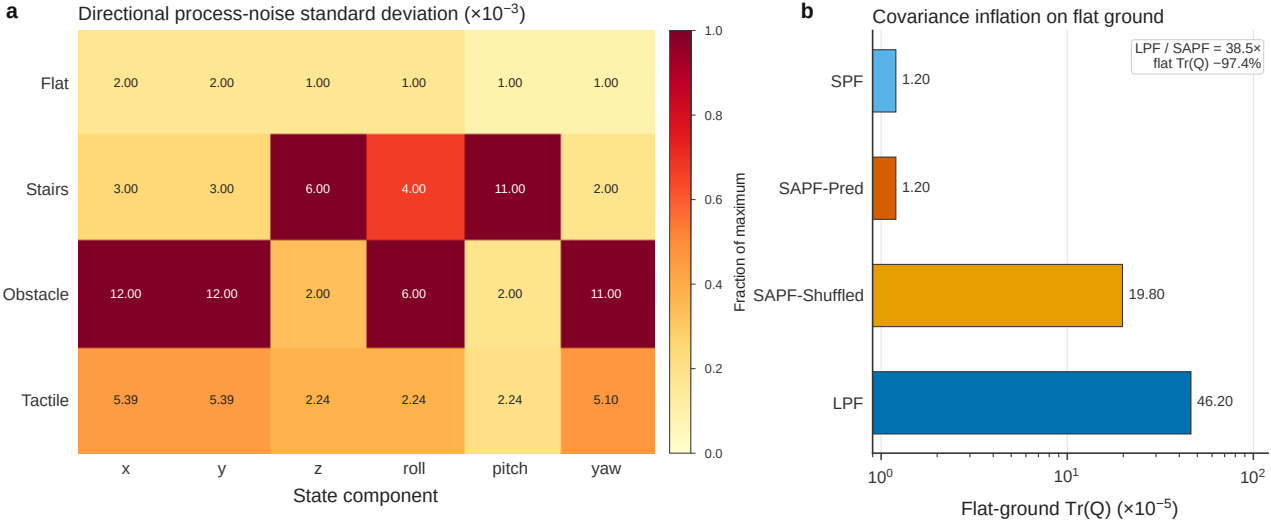
a
Directional process-noise standard deviation (×10⁻³)
Flat
Stairs
Obstacle
Tactile
2.00 2.00 1.00 1.00 1.00 1.00
3.00 3.00 6.00 4.00 11.00 2.00
12.00 12.00 2.00 6.00 2.00 11.00
5.39 5.39 2.24 2.24 2.24 5.10
x
y
z
roll
pitch
yaw
State component
Fraction of maximum
1.0
0.8
0.6
0.4
0.2
0.0
b
Covariance inflation on flat ground
LPF / SAPF = 38.5×
flat Tr(Q) −97.4%
SPF
SAPF-Pred
SAPF-Shuffled
LPF
1.20
1.20
19.80
46.20
10⁰
10¹
10²
Flat-ground Tr(Q) (×10⁻⁵)